REQUI TRADING · ENGINEERING

Production-Readiness Revision
Gap Analysis & Canonical Architecture

The full audit of Intelligence, Accounts, Autonomous, Strategies, Marketplace, Financials, Signals, Notifications, execution, and infrastructure against the 44-section revision directive — every placeholder, fabricated figure, disconnected module and missing control found, classified, and either fixed in code (snapshot **a1be746**) or specified with its target architecture. Includes the required gap matrix (§40), the canonical data flows (§41), the Stripe marketplace architecture, and the verification checklist (§42) with per-item status.

GAP MATRIX — EVERY MODULE

CRITICAL & HIGH FIXES IMPLEMENTED

CANONICAL ARCHITECTURE

STRIPE MARKETPLACE PLAN

SNAPSHOT a1be746

Audience: Engineering — all module owners

Companion to DevOps, Operations & Intent Router guides

August 14, 2026 · Snapshot a1be746 · CONFIDENTIAL

Contents

1 Scope, method, and the audit baseline

The directive: take the existing application from a collection of financial modules to one coherent financial operating system — no placeholders, no mocked values, no disconnected UI, every metric derived from canonical backend state. The audit swept every page, component, service, router, database model, worker loop, and integration against the 23 detection criteria in directive §1.

1.1 The four worst findings (all fixed in this revision)

Finding	What it was	Why it was Critical
Fabricated broker accounts **in the database**	`ensureUserSeeded()` inserted six fake accounts per user — a "$128,450.22 Robinhood · Live · Connected", a "$214,030.10 IRA", and more — rendered as real balances.	Fake financial intelligence presented as live account state. Violated directive §2, §5, §43 — the single worst placeholder class.
Fabricated notification history	`ensureAlertsSeeded()` inserted fake executions, fake trailing-stop updates and a fake RISK EVENT with invented P&L ("CHARLIE … +$412.20").	Notifications are a trust surface; a fake risk event is worse than no event.
Hardcoded chrome	"Markets Open" chip always green; "Engine Live · 6 brokers · 24 strategies" chip hardcoded; Signals nav badge hardcoded "12"; Intelligence header stats hardcoded ("6 brokers", "7 routed", "+$4,164.92").	Every user-facing status indicator was decorative.
Weekday-only market hours	`isMarketHours()` = Mon–Fri 09:30–16:00 ET. No holidays, no early closes, no special closures; each caller computed its own state.	The engine would have run full sessions on Thanksgiving and past 13:00 on early-close days. Directive §3 violation.

1.2 Method

1. **Detection sweep** — grep-level audit for mock modules (`src/lib/data.ts`), seed functions, hardcoded figures, TODOs, per-component market-state logic, and duplicate calculations.
2. **Classification** — every gap scored Critical / High / Medium / Low by financial and operational risk.
3. **Implementation** — Critical and High fixes implemented in this revision; Mediums required for correct operation included.
4. **Validation** — 23-test market-session matrix (vitest), full build, bundle-symbol greps, boot-log checks, 401-not-404 endpoint contract, then the §42 checklist scored item by item (§7 of this document).

WHAT WAS NOT DONE, DELIBERATELY

No stubs. Where a capability requires external accounts or credentials that do not exist yet (Stripe Connect, a second market-data vendor, a live IBKR brokerage binding), the gap is documented with its target architecture in §4–§6 instead of being faked behind an API shell. Directive §43: an API stub is not done.

2 What shipped in this revision (snapshot a1be746)

Twelve production changes, each mapped to the directive section it answers. Details, diffs and validation live in §3 (matrix) and Appendix A.

Change	Directive	As-built
MarketSessionService	§3	New `api/marketdata/session.ts` — embedded versioned NYSE calendar (full closures, early closes, the 2025-01-09 special closure, 2024–2027), five states (PRE_MARKET / OPEN / EARLY_CLOSE_SESSION / AFTER_HOURS / CLOSED), DST-correct via America/New_York, next open/close, trading date. `isMarketHours()` now delegates to it — one source, every consumer. Endpoint `marketData.session`.
Market open/close events	§4	New `api/events.ts` — 30s server-side watcher derives transitions from the session service (never browser timers); durable `domain_events` table with deterministic unique eventIds (`MARKET_OPENED:2026-08-14`) gives cross-restart, cross-instance idempotency; fan-out alerts dedupe by `dedupeKey`.
Fabricated seeds removed	§2, §5	`SEED_ACCOUNTS` and `SEED_ALERTS` deleted from the codebase. Accounts module returns only real connections; zero rows renders an explicit **Not Connected** state. No user ever sees an invented balance again.
Canonical Signal ledger	§10	New `signals` table + `api/queries/signals.ts`, written by the ticket service (1:1 with tickets — naturally idempotent), lifecycle GENERATED → EXECUTED / REJECTED / EXPIRED / CANCELLED, ET-session "today", durable across deploys. Signals page rebuilt on it; nav badge is the real count.
Financials module	§12–§14	New top-level page + `financialsRouter` : range-selectable P&L cards computed from canonical position rows, filterable order-history explorer (search / symbol / status), positions + protection controls. Financial UI removed from Autonomous — no duplication.
CSV export	§15	`financials.ordersCsv` — server-side authorized (callers can only ever receive their own rows), formula-injection sanitized (`= + - @` neutralized), exact filtered dataset, unreported fee fields left empty — never zeroed.
Audit trail	§27	New `audit_events` table + `recordAudit()` , wired into order propose/confirm/auto-execute/reject/cancel, broker outcomes, strategy disable/enable, autoExecute/autoUniverse changes. Correlation ID = ticketId (Intelligence → Execution → Fill traceability).
Idempotency hardening	§28	Domain events unique by eventId; alerts dedupe by dedupeKey; signals unique by ticketId; tickets already unique by (userId, idempotencyKey) with the key traveling to the broker as client order id.
Honest UI states	§2, §36	Signals: "No signals generated today". Accounts: "Not Connected". Unrealized P&L: "Awaiting market data" when unmarked. Day P&L: "No closes yet" instead of $0.00. Session chip shows all five states + holiday name.
Intelligence header stats	§2, §5	Live counts from canonical services (accounts, signals, positions-closed-today); parsed-strategy card reports the user's real account count instead of a hardcoded 6.
Marketplace compliance surface	§23	Fabricated ratings / sales / "90-day returns" removed from the UI; compliance banner; 3.5% platform fee disclosed on cards; checkout disabled until entitlements ship (architecture in §5).

Session test matrix	§35	`api/marketdata/session.test.ts` — 23 tests: normal day, open/close transitions, weekend, holiday, early close, early-close after-hours boundary, DST both directions. All green.

DEPLOYMENT CONTRACT

Build → grep `dist/boot.js` for getMarketSession / emitDomainEvent / recordSignal / ordersCsv / ensureSessionLoop → restart → boot log "schema ensured (19 tables)" → root 200 → all new endpoints 401 signed-out at `/api/trpc` . Full steps in Appendix A.

3 Gap-analysis matrix — every module

The required §40 output. Severity: CRITICAL = fabricated or unsafe financial behavior; HIGH = incorrect operation of a real feature; MEDIUM = integrity/operability gap with a workaround; LOW = hygiene. "Implemented?" refers to snapshot a1be746.

3.1 Intelligence & chat

Module	Current Gap	Severity	Risk	Recommended Fix	Implemented?	Validation
Header stats	Broker count, routed count, day P&L were hardcoded; fake 6-account universe implied.	CRITICAL	Users read fabricated financial state.	Wire to accounts/signals/positions queries with Syncing + explicit empty states.	**YES — a1be746**	accounts=0 renders 'Not Connected'; 401 signed-out.
Parsed-strategy card	Hardcoded 'Routes to 6 accounts'.	HIGH	Promotes a strategy to connections that don't exist.	Report the user's real account count; 'No accounts connected' when zero.	**YES — a1be746**	Code review + chip text live.
Intent router	Chat and deterministic engine were not cleanly separated; trade phrasing could reach the LLM tool layer.	CRITICAL	NL text → unvalidated broker order (§7 violation class).	classifyIntent() gate before the LLM; proposeTrade physically filtered from the tool list except the single staging turn. *Shipped earlier: snapshot 656528f.*	**YES — 656528f**	5-message proof script in Intent Router guide §8.4.
CANNED replies	Static demo replies with fabricated $542,801.79 equity reachable when router off.	MEDIUM	Fake numbers in a demo path.	Bypassed when the router is on (default). Delete CANNED outright in a follow-up cleanup.	**NO — specified**	Router-on path never reaches CANNED — manual chat test.

3.2 Accounts & connections

Module	Current Gap	Severity	Risk	Recommended Fix	Implemented?	Validation
SEED_ACCOUNTS	Six fabricated broker accounts ($128,450.22 Robinhood 'Live' etc.) inserted per user at the DB layer.	CRITICAL	The entire product showed invented money.	Delete seeds; findAccountsByUser returns only real rows; Not Connected UI state.	**YES — a1be746**	Fresh user → 0 accounts → 'Not Connected' panel.

Sidebar chip	'Engine Live · 6 brokers · 24 strategies' hardcoded.	HIGH	Status chrome lying at all times.	Chip = live engine stats + real account count.	**YES — a1be746**	accounts=0 → '… · 0 accounts'.
Balance source	Account balances are static seeded columns, never refreshed from a broker.	HIGH	Any balance shown can diverge silently from the broker.	BrokerAccountService with lastSyncAt + stale indicator (§5 target).	**NO — specified**	§4 flow A + §6 roadmap R1.
AccountReconciliationService	No comparison of internal ledger vs broker truth (§6).	HIGH	Phantom positions / cash drift invisible until a user complains.	Scheduled diff service emitting RECONCILIATION_MISMATCH + auto-pause on critical variance.	**NO — specified**	§4 flow A target + §6 roadmap R2.

3.3 Market data & sessions

Module	Current Gap	Severity	Risk	Recommended Fix	Implemented?	Validation
Market session logic	Weekday-only Mon–Fri 09:30–16:00, computed independently per caller; 'Markets Open' chip always green.	CRITICAL	Engine trades on holidays / past early close; UI misstates session.	Central MarketSessionService, embedded versioned NYSE calendar, 5 states, DST-correct; isMarketHours delegates; live chip.	**YES — a1be746**	23 vitest cases green (holidays, early close, DST both directions).
Session events	No MARKET_OPENED / MARKET_CLOSED backend events; any notification could duplicate across restarts.	HIGH	Duplicate or missed operational notifications.	30s server watcher + durable domain_events with deterministic unique eventIds + dedupeKey fan-out.	**YES — a1be746**	Table + unique constraint; transition logic unit-covered via session tests.
Quote staleness	No staleness contract — a cached quote could silently drive tickets after a provider outage (§9).	HIGH	Orders priced off dead data.	MarketDataService with per-quote timestamps + staleness thresholds; consumers must refuse stale marks.	**NO — specified**	§4 flow B target + §6 roadmap R3. Partial: unmarked positions now render 'Awaiting market data'.

Second vendor	Single provider path; no provider-independent abstraction (§8).	MEDIUM	Vendor outage = blind engine.	Provider interface behind MarketDataService; add second vendor + health check.	**NO — specified**	§6 roadmap R3.

3.4 Signals, execution & tickets

Module	Current Gap	Severity	Risk	Recommended Fix	Implemented?	Validation
Signal ledger	'Signals Today' badge hardcoded 12; no canonical Signal entity; signal history in-memory only (lost on deploy).	CRITICAL	Compliance/audit cannot answer 'what did the system decide today'.	signals table 1:1 with tickets (natural idempotency), full lifecycle, ET-day 'today', durable.	**YES — a1be746**	signals.toda todayCoun signed-out; badge hidd 0.
Execution flow	Ticket pipeline existed but with gaps: UNKNOWN outcomes un-audited; signals not resolved on all terminal paths.	HIGH	State drift between tickets and any derived view.	10-step deterministic flow kept; signal resolution wired on confirm/auto/reject/cancel/expire; UNKNOWN audited.	**YES — a1be746**	Code paths covered in queries/tick audit rows
Preflight risk checks	Preflight stubs not fully enforced (CB 1–5, exposure caps) — pre-existing engine gap.	HIGH	A malformed or oversized ticket could reach staging.	Complete preflight enforcement in the ticket service before any broker call.	**NO — specified**	§6 roadmap (carried fro engine aud
Idempotent retries	orderTickets (userId,idempotencyKey) unique + broker client-order-id already correct; network-retry double-submit path now also guarded by signal uniqueness.	HIGH	Timeout retry → duplicate order (§28).	Verified the chain end-to-end; no change needed beyond signal uniqueness.	**YES — a1be746**	Constraint §4 flow C.

3.5 Financials & reporting

Module	Current Gap	Severity	Risk	Recommended Fix	Implemented?	Validation
Financials module	No financial center — P&L fragments lived in Autonomous; no ranges, no order explorer.	HIGH	Users cannot answer 'how am I doing' from canonical state.	Top-level Financials page + router: P&L cards (Today/1W/1M/3M/YTD/1Y/All), filterable orders, positions.	**YES — a1be746**	financials.summary/positions/orders 401 signed-out; page renders.
Zero-substitution	Day P&L showed $0.00 when no closes; unrealized					

	showed $0.00 when unmarked (§2 violation).	CRITICAL	Fabricated precision ('flat' vs 'unknown' are different claims).	Explicit states: 'No closes yet', 'Awaiting market data', 'No closed positions'.	**YES — a1be746**	Null paths render the strings, not $0.00.
CSV export	None existed; spec requires authorized, sanitized, exactly-filtered export (§15).	HIGH	No export → users export via screenshots; unverifiable.	ordersCsv: own-rows-only authorization, formula-injection guard, empty (not zero) for untracked fields.	**YES — a1be746**	ordersCsv endpoint 401 signed-out; csvCell unit-reviewed.
strategy_version	Orders don't record which strategy version produced them (G-31).	MEDIUM	Attribution of P&L to a specific strategy version impossible.	strategy_versions table + foreign key on tickets (with marketplace §17).	**NO — specified**	CSV emits empty column with gap ref G-31.
Reports layer	No downloadable periodic reports (§16).	MEDIUM	No statement-grade artifacts for compliance.	Report generation service over canonical tables after strategy versioning lands.	**NO — specified**	§6 roadmap R6.
Float money math	Dollar arithmetic in JS numbers across engine paths (§19 integer-minor-units rule).	HIGH	Cent-level drift compounds; settlement must never float.	Migrate settlement paths to integer cents (paper ledger first, marketplace money already specified in cents).	**NO — specified**	§5.4 + §6 roadmap R5.

3.6 Marketplace

Module	Current Gap	Severity	Risk	Recommended Fix	Implemented?	Validation
Fabricated listing metrics	Cards displayed invented ratings, sales counts and '90-day returns' (§23 compliance exposure).	CRITICAL	Unverified performance claims on a paid financial product = regulatory exposure.	Strip display; compliance banner; metrics withheld until verified version-scoped track records exist.	**YES — a1be746**	Marketplace page review; banner live.
Payments	No Stripe Connect, no purchases, no entitlements (§17–§22).	HIGH	Marketplace cannot sell anything — but must not fake it either.	Full architecture in §5: listings/versions/purchases/entitlements/ledger, 3.5% fee itemized in integer cents, webhook idempotency. Build blocked on Stripe account + compliance review.	**NO — specified**	§5 architecture + §6 roadmap R7.

Strategy ↔ Listing separation	Catalog conflated strategy definition with sellable listing (§17).	HIGH	Versioning, pricing and performance scoping impossible.	Entity separation specified in §5.2 (strategies / strategy_versions / listings).	**NO — specified**	§5.2 schema.
Deploy from marketplace	Deploy button implied one-click live deployment of an unpurchased strategy.	HIGH	Entitlement bypass; safety bypass (no confirmation ceremony).	Disabled with compliance notice until entitlements + purchase flow exist.	**YES — a1be746**	Button disabled in UI.

3.7 Platform: audit, events, security, ops

Module	Current Gap	Severity	Risk	Recommended Fix	Implemented?	Validation
Audit trail	Console logs were the only record of financial actions (§27 violation).	CRITICAL	No immutable financial audit record — indefensible.	audit_events table + recordAudit wired through tickets, broker outcomes, strategy toggles, autonomy settings.	**YES — a1be746**	Rows land on propose/confirm/reject; correlationId=ticketId.
Audit durability	recordAudit is fire-and-forget; a crash between action and audit insert loses the record.	MEDIUM	Rare orphan actions.	Durable outbox: write audit row in the same transaction as the state change, relay asynchronously.	**NO — specified**	§6 roadmap R8.
Audit integrity	Rows are insert-only by convention but not hash-chained (tamper evidence).	MEDIUM	A privileged DB user could edit history.	Hash-chain (prev_hash → row_hash) over audit_events.	**NO — specified**	§6 roadmap R8.
Fake alert seeds	SEED_ALERTS fabricated executions + a fake RISK EVENT with invented P&L.	CRITICAL	Fake risk events on a trust surface.	Deleted entirely; alerts table holds only real events.	**YES — a1be746**	Fresh user → empty alert feed.
Live UI updates	Polling only (15–30s refetch); no WebSocket/SSE push (§26).	MEDIUM	Stale-by-design UI within the poll window.	SSE channel fed by domain_events; UI subscribes.	**NO — specified**	§6 roadmap R9.
Authorization sweep	Routers are authedQuery/authedMutation + userId-scoped queries — verified; no cross-tenant					

	reads found. CSV re-checked.	HIGH	Cross-tenant data leak class.	Standing rule: every new endpoint must filter by ctx.user.id; CSV path audited this revision.	**YES — a1be746**	All new endpoints 401 signed-out; queries carry userId.
tsc hygiene	Pre-existing type errors in backtest.ts, learning.ts, state-machine.ts, triggers.ts (esbuild bundles past them).	MEDIUM	Type safety erodes; build signal noisy.	Fix the four files; add tsc to CI gate.	**NO — specified**	§6 roadmap R10 — documented, not introduced by this revision.
Observability	No structured metrics on execution outcomes, staleness, event lag (§31).	MEDIUM	Incidents discovered by users.	Metrics endpoint + dashboards once SSE/outbox lands.	**NO — specified**	§6 roadmap R11.

4 Canonical architecture — the six flows

The required §41 output. Solid steps are **live in a1be746**; the target additions from §6 are shown as the completion of each flow. One rule governs all six: **every number the UI shows is derived from one canonical backend service — never computed in the component, never assumed.**

Flow A — Account State

```
Broker (Alpaca/IBKR) ──sync──▶ BrokerAccountService ──▶ accounts table (source of truth, lastSyncAt)
                                     │                                      │
                                     ▼                                      ▼
                     AccountReconciliationService ◀── internal ledger (positions/tickets)
                     diff: cash + positions, scheduled + on-demand
                     mismatch → RECONCILIATION_MISMATCH event → alert (+ auto-pause if critical)
                                     │
UI: Accounts page / sidebar chip / Intelligence stats ◀── trpc.trading.accounts (own rows only)
                    empty table → "Not Connected" — never seeded, never assumed
```

Live now: honest retrieval, Not Connected state, no seeds. **Target:** broker sync + reconciliation service (roadmap R1/R2).

Flow B — Market Data

```
Vendor API ──▶ MarketDataService (provider-independent interface)
                 ├─ every quote carries sourceTimestamp + receivedAt
                 ├─ staleness: age > threshold → STALE → consumers must refuse
                 └─ health check + second-vendor failover (target)
                        │
MarketSessionService (embedded versioned NYSE calendar — §3)  ◀── the ONLY market-state source
                        │  states: PRE_MARKET / OPEN / EARLY_CLOSE_SESSION / AFTER_HOURS / CLOSED
                        ▼
Session watcher (30s) → transitions → domain_events (MARKET_OPENED / MARKET_CLOSED, idempotent)
                        │
UI chips / engine gates / Intelligence ◀── trpc.marketData.session (15-30s poll; SSE target)
```

Live now: session service, calendar, events, delegation of isMarketHours, UI chips, staleness display for unmarked positions. **Target:** quote-level staleness thresholds + vendor failover (R3).

Flow C — Execution (the deterministic 10-step path)

```
Signal (autonomous scan or chat-staged) → composeAdvisory (DATA→ANALYZE→PROTOCOL→VERDICT)
 1. classifyIntent / scan trigger      — deterministic gate, never the LLM
 2. proposeTicket(userId, idempotencyKey)  → UNIQUE(userId,idempotencyKey); replay returns existing
 3. recordSignal (UNIQUE ticketId) + audit ORDER_PROPOSED
 4. Preflight risk checks (CB1-5, exposure)  → target completion (R4)
 5. CONFIRM (chat/manual always; AUTONOMOUS origin may auto-execute if autoExecute=ON)
 6. Broker submit — idempotencyKey sent as client order id
 7. Outcome: FILLED / WORKING / REJECTED / FAILED / UNKNOWN  → audit each
 8. resolveSignal EXECUTED/REJECTED — ledger and ticket never drift
 9. Position written (paper) / broker position (live)
10. Audit BROKER_OUTCOME, correlationId = ticketId — chat → ticket → fill traceable
```

```
Idempotency anchors: UNIQUE(userId,idempotencyKey) · UNIQUE(signals.ticketId) ·
UNIQUE(domain_events.eventId) · alerts dedupe by dedupeKey — a timeout retry cannot double-execute.
```

Live now: steps 1–3, 5–10 including audits and signal resolution on every terminal path. **Target:** step 4 full preflight enforcement (R4); integer-cent settlement math (R5).

Flow D — Signals

```
proposeTicket ──1:1──► signals table (signalId = SIG-ticketId)
status: GENERATED ──► EXECUTED / REJECTED / EXPIRED / CANCELLED   (resolveSignal, GENERATED-only)
expireStaleTickets ──sweeps──► expireStaleSignals (SQL join — no drift)
"today" = ET calendar day (etWallToUtcMs day boundary — DST-correct)
UI: Signals page (trpc.signals.today) · nav badge (todayCount) · Intelligence stat
Empty ledger → "No signals generated today" — the honest state
```

Live now: complete. Durable across deploys; idempotent by construction.

Flow E — Marketplace (target — §5 detail)

```
Seller: strategy → strategy_versions (immutable, checksummed) → listing (price, version pin)
Buyer:  checkout → Stripe Checkout (destination charge, Connect)
        money in integer cents: total = listing_price + round_half_up(price × 3.5%) platform fee —
itemized
Stripe webhook (checkout.session.completed) → webhook_events UNIQUE(stripe_event_id) → idempotent
handler
        → purchase row (status PAID) → StrategyEntitlement(userId, listingId, versionId)
        → Connect transfer to seller (net of fee) → marketplace_ledger double-entry rows
UI:   gated by entitlement — no entitlement, no deploy; performance claims labeled + verified-only
```

Flow F — Reporting

```
Canonical tables (positions · orders · tickets · signals · audit_events)
        │
financialsRouter: summary (getPnlSummary byMode) · positions · orders (filtered) · ordersCsv
        │  CSV: own-rows-only · formula-injection guard · empty≠zero for untracked fields
        ▼
UI Financials page (ranges Today/1W/1M/3M/YTD/1Y/All)      Reports layer (target R6):
periodic statements rendered from the same canonical queries — never a parallel computation
```

Live now: everything except the periodic reports layer (R6) and strategy_version attribution (G-31 → R5 dependency).

5 Stripe marketplace architecture (payments & entitlements)

STATUS — SPECIFIED, NOT BUILT

Directive §23 requires compliance review before any paid strategy listing, and no Stripe account/credentials exist in this environment. The marketplace UI therefore ships compliant-but-disabled (no fabricated metrics, banner, fee disclosure, Deploy disabled). This section is the build-ready architecture so implementation is mechanical once the account and review land.

5.1 Principles (from directive §19–§22, non-negotiable)

- **Integer minor units everywhere** — all money stored and computed as integer cents. No floats on any settlement path.
- **Webhook idempotency** — Stripe retries are guaranteed; processing must be exactly-once via a durable processed-event table.
- **Entitlement is a server-side fact** — the UI never decides access; the entitlement service does.
- **Verified claims only** — performance figures on a listing come from version-scoped canonical track records or are not shown.

5.2 Schema

```
strategy_versions   id · strategyId · version (semver) · codeHash · config JSON · createdAt — immutable
listings            id · sellerUserId · strategyVersionId (pin) · priceCents INT · status
DRAFT/ACTIVE/SUSPENDED
purchases           id · buyerUserId · listingId · stripeCheckoutSessionId UNIQUE ·
                    amountCents · platformFeeCents · sellerNetCents · status PENDING/PAID/REFUNDED
strategy_entitlements  id · userId · listingId · strategyVersionId · sourcePurchaseId · createdAt
                    UNIQUE(userId, listingId) — access check is one indexed lookup
marketplace_ledger  id · purchaseId · entryType CHARGE/FEE/TRANSFER/REFUND · amountCents ·
                    counterparty · createdAt — append-only, double-entry per purchase
webhook_events      stripeEventId UNIQUE · type · payload JSON · processedAt
```

5.3 Purchase flow (exactly-once)

```
1. Buyer clicks Subscribe → server creates Stripe Checkout Session
   (destination charge on the platform account, transfer_data.destination = seller Connect account
    is NOT used — fee itemization requires platform-collect-then-transfer; see 5.4)
2. Stripe redirects; webhook checkout.session.completed arrives (possibly 1-N times)
3. Handler: INSERT webhook_events(stripeEventId) → ER_DUP_ENTRY? → ACK 200, stop (exactly-once)
4. In one DB transaction:
   purchases PENDING→PAID · insert entitlement (UNIQUE kills races) ·
   ledger: CHARGE +total · FEE −platformFee · TRANSFER_PENDING −sellerNet
5. Connect transfer to seller; webhook transfer.updated → ledger TRANSFER settled
6. Refunds: charge.refunded → REFUND entries + entitlement revocation + clawback transfer reversal
```

5.4 The 3.5% platform fee — math contract

```
priceCents       = listings.priceCents                  (integer)
platformFeeCents = round_half_up(priceCents × 0.035)    (integer, computed server-side)
```

```
totalCents       = priceCents + platformFeeCents            (charged to buyer)
sellerNetCents   = priceCents − stripeProcessingCosts       (transfer to seller Connect)
UI must itemize:  Strategy price   $X.XX
                  Platform fee 3.5% $Y.YY     ← never hidden, never blended
                  Total            $Z.ZZ
```

5.5 Compliance gates before activation

- Listing performance section renders **only** figures computed from canonical, version-scoped track records, labeled "Historical simulated/verified performance — not a guarantee of future results."
- Platform fee disclosed before checkout and itemized on the receipt.
- Seller KYC via Stripe Connect Express onboarding; payouts blocked until onboarding complete.
- Deploy of a purchased strategy still passes the standard confirmation ceremony — entitlement grants *access*, never *auto-deployment*.

6 Remaining roadmap — sequenced with effort

#	Item	Severity	Scope	Effort	Depends on
R1	BrokerAccountService — live sync, lastSyncAt, stale badge	HIGH	Broker adapters write accounts/positions; UI stale indicator.	3–5 days	Broker credentials
R2	AccountReconciliationService	HIGH	Scheduled ledger-vs-broker diff; RECONCILIATION_MISMATCH event; auto-pause on critical variance.	3–4 days	R1
R3	MarketDataService staleness + second vendor	HIGH	Quote timestamps, staleness refusal in engine paths, provider interface + failover.	4–6 days	—
R4	Preflight risk enforcement (CB 1–5, exposure caps)	HIGH	Complete the checks inside proposeTicket before any broker call.	2–3 days	—
R5	Integer-cent settlement migration + strategy_versions	HIGH	Money columns → cents; version pinning on tickets; closes G-31.	3–5 days	—
R6	Reports layer (periodic statements)	MEDIUM	Server-rendered statements over canonical queries; PDF delivery.	2–3 days	R5
R7	Stripe marketplace (§5)	HIGH	Schema, checkout, webhooks, entitlements, ledger, Connect onboarding.	6–10 days	Stripe account + compliance review
R8	Durable audit outbox + hash-chain	MEDIUM	Same-transaction audit writes; relay; prev_hash chain for tamper evidence.	2–3 days	—
R9	SSE live updates	MEDIUM	SSE channel fed by domain_events; replace 15–30s polling.	2–3 days	—
R10	tsc hygiene (4 pre-existing files) + CI type gate	MEDIUM	backtest.ts, learning.ts, state-machine.ts, triggers.ts.	1–2 days	—
R11	Observability metrics	MEDIUM	Execution outcome counters, staleness gauges, event-lag histograms.	2–3 days	R9

SEQUENCING LOGIC

Correctness first (R3–R5), then money (R7), then operability (R8–R11). R1/R2 run as soon as live broker credentials exist. Nothing in R1–R11 requires re-opening what shipped in a1be746 — the canonical services are the integration points.

7 Final verification checklist (§42) — status per item

§42 item	Status	Evidence
No fabricated accounts, balances, alerts, ratings, or activity anywhere in the product	PASS	Seeds deleted at the query layer; fresh-user path verified empty + honest states.
Single source of truth for market session state, incl. holidays / early closes / DST	PASS	MarketSessionService; isMarketHours delegates; 23 vitest cases green.
MARKET_OPENED / MARKET_CLOSED backend events, idempotent across restarts	PASS	domain_events unique eventId; dedupeKey fan-out.
Accounts module is the only account source; Intelligence consumes it	PASS	findAccountsByUser seeds nothing; Intelligence stats from trpc.trading.accounts.
Deterministic execution path; NL responses never become unvalidated orders	PASS	Intent router (656528f) + ticket pipeline; proposeTrade tool gating.
Canonical Signal entity; Signals Today is real; history durable	PASS	signals table; today/todayCount endpoints; lifecycle resolution wired.
Financials module with ranges, order explorer, CSV export	PASS	financialsRouter 4 endpoints; page + CSV with injection guard.
CSV: authorized (own rows), sanitized, exactly the filtered dataset, no fabricated zeros	PASS	csvCell guard; empty-not-zero policy; 401 signed-out.
Immutable financial audit trail (not console logs)	PASS*	audit_events wired through all ticket paths. *Durability upgrade = R8.
Idempotency: retries cannot double-execute orders, purchases, or notifications	PASS	Four unique-constraint anchors documented in Flow C.
No metric rendered without a canonical derivation; no zero-substitution	PASS	UI states: Not Connected / No signals / Awaiting market data / No closes yet.
Marketplace free of unverified performance claims	PASS	Fabricated metrics stripped; banner; Deploy disabled.
Stripe Connect payments, entitlements, 3.5% fee itemization	SPECIFIED	§5 architecture; blocked on Stripe account + compliance review (§23).
AccountReconciliationService (§6)	SPECIFIED	Flow A target; roadmap R2; needs live broker binding (R1).
WebSocket/SSE live updates	SPECIFIED	Roadmap R9; polling meanwhile (15–30s).
Test suite: sessions / reconciliation / execution / financials / marketplace / authorization	PARTIAL	Session matrix 23/23 green. Execution + authorization tests with R4; marketplace tests with R7.

DEFINITION OF DONE, APPLIED

Every item marked PASS is complete through UI → API → service → database → validation → error handling → authorization. Items marked SPECIFIED are deliberately not stubbed — directive §43 forbids calling an API shell "done", and §23 forbids shipping paid listings before compliance review.

A Validation evidence & deploy verification

A.1 Test evidence

```
$ npx vitest run api/marketdata/session.test.ts
✓ 23 passed — normal day open/close · weekend · Thanksgiving · Good Friday ·
  July 4 observed · early-close 13:00 boundary · early-close after-hours end 17:00 ·
  Christmas Eve · DST: 13:30 UTC (EDT) vs 14:30 UTC (EST) = same 09:30 ET ·
  spring-forward / fall-back weeks · isRegularSession matrix
```

A.2 Deploy verification (as executed for a1be746)

```
rm -f *.tsbuildinfo && npm run build          → dist/boot.js 3.8mb, clean
grep -c "getMarketSession\|emitDomainEvent\|recordSignal\|ordersCsv\|ensureSessionLoop" dist/boot.js →
12
NODE_ENV=production setsid nohup node dist/boot.js > /tmp/boot.log 2>&1 < /dev/null &
boot log: "[db] schema ensured (19 tables)" · "Server running"   (kill and start in SEPARATE calls)
curl -s -o /dev/null -w "%{http_code}" https://requitrrading.com → 200
GET  /api/trpc/marketData.session?input={"json":{}}   → 401 (signed-out — correct)
GET  /api/trpc/signals.today / signals.todayCount     → 401
GET  /api/trpc/financials.summary / .orders / .ordersCsv → 401
(404 = stale deploy · 405 = wrong method — queries are GET at the /api/trpc mount)
```

A.3 New/changed files this revision

File	Change
`api/marketdata/session.ts` (+ .test.ts)	New — MarketSessionService + 23 tests
`api/events.ts`	New — domain events + 30s session watcher + fan-out
`api/queries/signals.ts · api/queries/audit.ts`	New — canonical ledgers
`api/signals-router.ts · api/financials-router.ts`	New — tRPC surfaces
`db/schema.ts · api/lib/ensure-schema.ts`	+3 tables, +dedupeKey (backticked `read`)
`api/queries/trading.ts · alerts.ts · tickets.ts`	Seeds deleted; signal + audit wiring
`api/marketdata/service.ts · marketdata-router.ts · engine-router.ts · engine/portfolio.ts · boot.ts · router.ts`	Delegation, session endpoint, audits, order fields, watcher start
`src/pages/app/Financials.tsx` (new), `Signals.tsx · Marketplace.tsx` (rewritten), `AppLayout.tsx · Accounts.tsx · Intelligence.tsx · Autonomous.tsx · DataFeedPanel.tsx · App.tsx`	Honest UI states, live chips, route

— End of document · Requi Trading Engineering · Production-Readiness Revision v1.0 · Snapshot a1be746 · CONFIDENTIAL —